Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

REGISTER NUMBER: 23MIS0586

NAME: PRIYAN V

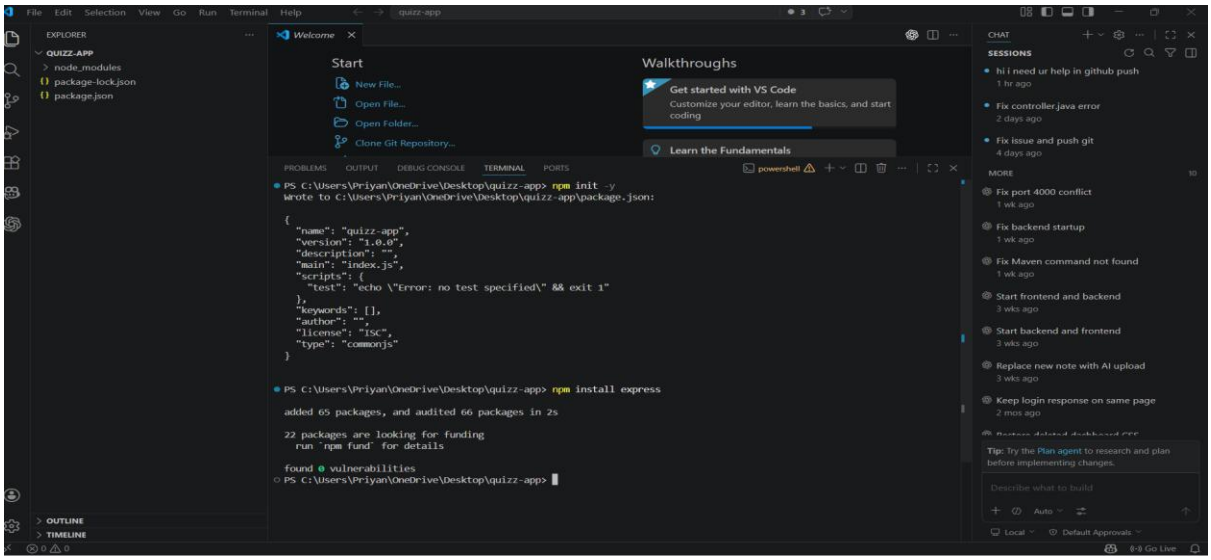
COURSE TITLE: AGILE DEVELOPMENT PROCESS AND DEVOPS

COURSE CODE: ISWE406L SLOT: C2+TC2

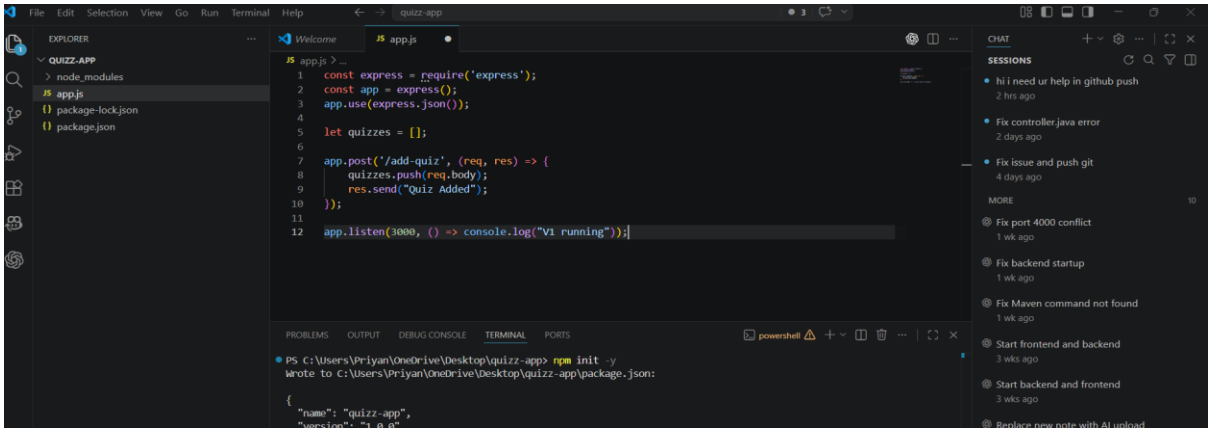
AGILE COURSE PROJECT

QUIZZ-APP

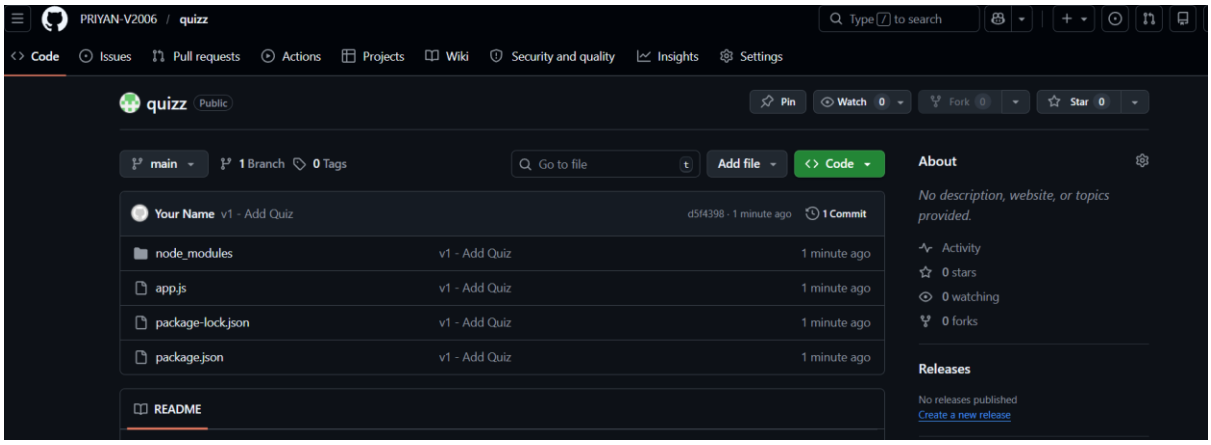
CREATE PROJECT IN (VS CODE)



V1 CODE (Add Quiz – Create Quiz)



STEP 2: GitHub Setup



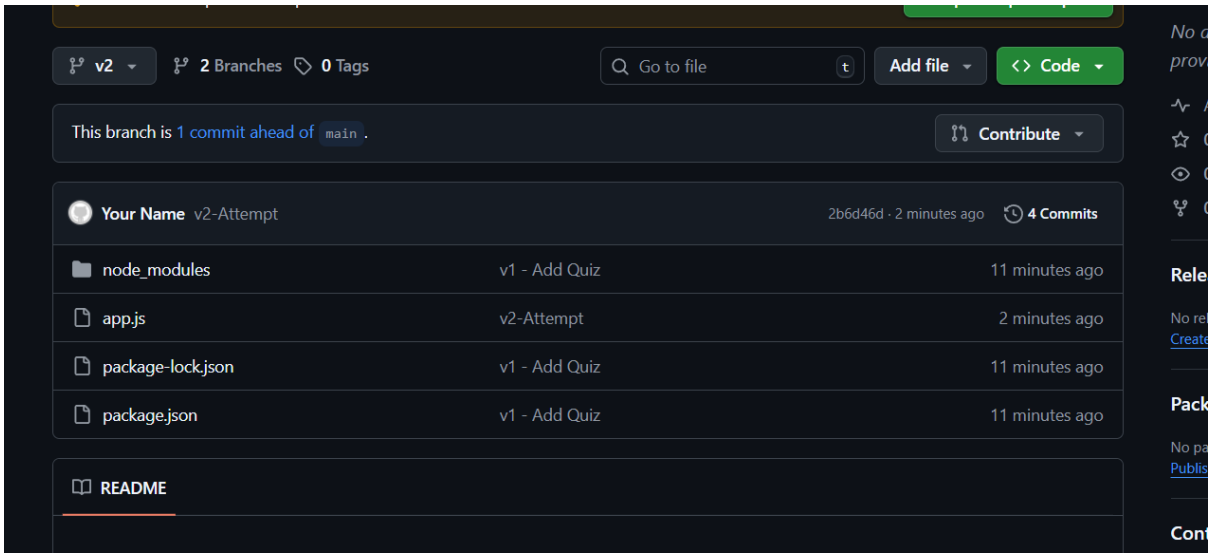
STEP 3: CREATE v2 (ATTEMPT QUIZ)

Update app.js

```
const express = require('express');
const app = express();
app.use(express.json());

let quizzes = [];

app.post('/add-quiz', (req, res) => {
    quizzes.push(req.body);
    res.send("Quiz Added");
});

app.post('/attempt', (req, res) => {
    res.send("Quiz Attempted");
});

app.listen(3000, () => console.log("V1 running"));
```

Commit

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> git checkout -b v2
Switched to a new branch 'v2'
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> git add .
>> git commit -m "v2 - Attempt Quiz"
>> git push origin v2
On branch v2
nothing to commit, working tree clean
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 429 bytes | 429.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/PRIYAN-V2006/quizz.git
   d5f4398..2b6d46d  v2 -> v2
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```



STEP 4: CREATE v3 (Score)

Add feature:

```
EXPLORER
QUIZZ-APP
node_modules
app.js
package-lock.json
package.json
```

```
app.js > ...
const express = require('express');
const app = express();
app.use(express.json());

let quizzes = [];

app.post('/add-quiz', (req, res) => {
    quizzes.push(req.body);
    res.send("Quiz Added");
});
app.post('/attempt', (req, res) => {
    res.send("Quiz Attempted");
});
app.get('/score', (req, res) => {
    res.send("Score: 10/10");
});

app.listen(3000, () => console.log("V1 running"));
```

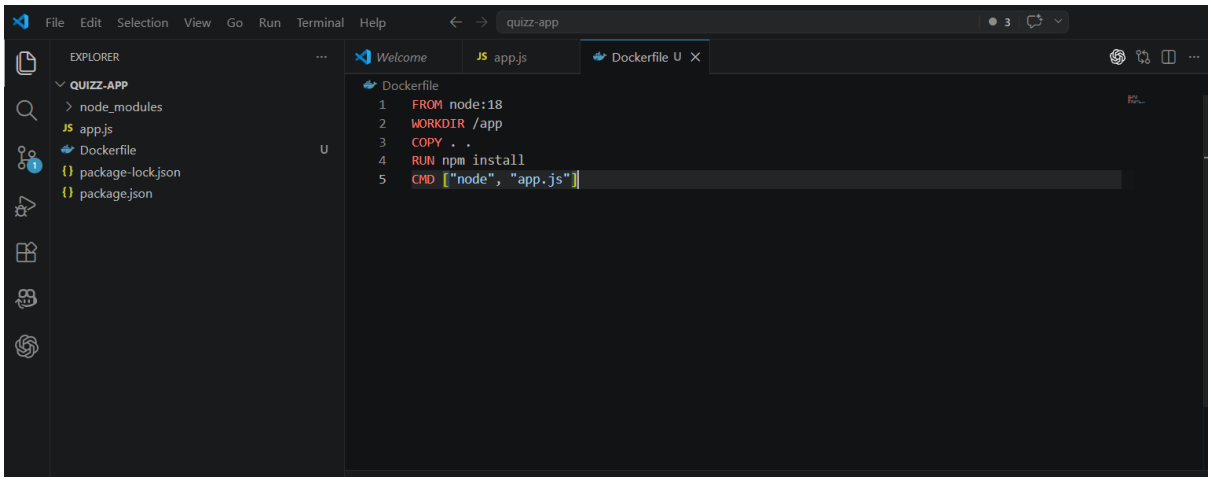
Commit:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> git checkout -b v3
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> git commit -am "v3 - Score feature"
>> git push origin v3
[v3 8000fd4] v3 - Score feature
 1 file changed, 3 insertions(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 407 bytes | 407.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'v3' on GitHub by visiting:
remote:      https://github.com/PRIYAN-V2006/quizz/pull/new/v3
remote:
To https://github.com/PRIYAN-V2006/quizz.git
 * [new branch]      v3 -> v3
Your Name (7 minutes ago)    Ln 18,
```

```
v3    3 Branches    0 Tags    Go to file t    Add file    Code

This branch is 2 commits ahead of main.    Contribute

Your Name  v3 - Score feature    8000fd4 · now    5 Commits

node_modules    v1 - Add Quiz    14 minutes ago
app.js          v3 - Score feature    now
package-lock.json    v1 - Add Quiz    14 minutes ago
package.json    v1 - Add Quiz    14 minutes ago
```

STEP 5: DOCKER SETUP

Create Dockerfile :



STEP 6: BUILD DOCKER IMAGES

V1:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker build -t quiz-app:v1 .
[+] Building 7.2s (10/10) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.1s
 => => transferring dockerfile: 114B                                                0.0s
 => [internal] load metadata for docker.io/library/node:18                          2.5s
 => [auth] library/node:pull token for registry-1.docker.io                         0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/4] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => [internal] load build context                                                   0.6s
 => => transferring context: 3.16MB                                                 0.6s
 => CACHED [2/4] WORKDIR /app                                                       0.0s
 => [3/4] COPY . .                                                                  0.5s
 => [4/4] RUN npm install                                                           2.4s
 => exporting to image                                                              0.9s
 => => exporting layers                                                             0.3s
 => => exporting manifest sha256:b382eefb74ad0d1ebf87f8fa6d9a60405ad33060fee79d320b252f7ab8298  0.0s
 => => exporting config sha256:321b3d6d0fd64e621ee889625c77909661c4a04b14e9c4be80dfebccbb92c8b  0.0s
 => => exporting attestation manifest sha256:c27485b0ca339a1bb0878824572a86f779439affa809be757  0.0s
 => => exporting manifest list sha256:9e4c3eb0ccac110e307af86dff127b94e061b5a0e7228780e76c9e51  0.0s
 => => naming to docker.io/library/quiz-app:v1                                      0.0s
 => => unpacking to docker.io/library/quiz-app:v1                                   0.5s
```

V2:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker build -t quiz-app:v2 .
[+] Building 1.6s (9/9) FINISHED                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 114B                                                  0.0s
 => [internal] load metadata for docker.io/library/node:18                            1.2s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/4] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => [internal] load build context                                                     0.1s
 => => transferring context: 100.08kB                                                 0.1s
 => CACHED [2/4] WORKDIR /app                                                         0.0s
 => CACHED [3/4] COPY . .                                                             0.0s
 => CACHED [4/4] RUN npm install                                                      0.0s
 => exporting to image                                                                0.1s
 => => exporting layers                                                               0.0s
 => => exporting manifest sha256:b382eefb74ad0d1ebf87f8fa6d9a60405ad33060fee79d320b252f7ab8298  0.0s
 => => exporting config sha256:321b3d6d0fd64e621ee889625c77909661c4a04b14e9c4be80dfebccbb92c8b  0.0s
 => => exporting attestation manifest sha256:1fa804fd34a5bdd330189e9bf1a0a098b56e316028cc2e7fc  0.0s
 => => exporting manifest list sha256:989a44dd0ed9f2b5342fff1e02ea5121dca235c2e4f85a7ccf05f855  0.0s
 => => naming to docker.io/library/quiz-app:v2                                        0.0s
 => => unpacking to docker.io/library/quiz-app:v2                                     0.0s
```

V3:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker build -t quiz-app:v3 .
[+] Building 1.0s (9/9) FINISHED                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 114B                                                  0.0s
 => [internal] load metadata for docker.io/library/node:18                            0.5s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/4] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4  0.0s
 => [internal] load build context                                                     0.1s
 => => transferring context: 100.08kB                                                 0.1s
 => CACHED [2/4] WORKDIR /app                                                         0.0s
 => CACHED [3/4] COPY . .                                                             0.0s
 => CACHED [4/4] RUN npm install                                                      0.0s
 => exporting to image                                                                0.1s
 => => exporting layers                                                               0.0s
 => => exporting manifest sha256:b382eefb74ad0d1ebf87f8fa6d9a60405ad33060fee79d320b252f7ab8298  0.0s
 => => exporting config sha256:321b3d6d0fd64e621ee889625c77909661c4a04b14e9c4be80dfebccbb92c8b  0.0s
 => => exporting attestation manifest sha256:d2c65297125ae610516234027d593b4d81700903a9c9aa3a7  0.0s
 => => exporting manifest list sha256:6994c127443a1a63f2fdc70d19ea317ead5101b69c8ffa2aced74bcf  0.0s
 => => naming to docker.io/library/quiz-app:v3                                        0.0s
 => => unpacking to docker.io/library/quiz-app:v3                                     0.0s
```

Verify docker images

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker images
                                                          i Info →   U  In Use
IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
quiz-app:v1    9e4c3eb0ccac       1.58GB          397MB
quiz-app:v2    989a44dd0ed9       1.58GB          397MB
quiz-app:v3    6994c127443a       1.58GB          397MB
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```

STEP 7: PUSH TO DOCKER HUB

Docker login

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker login
Authenticating with existing credentials... [Username: priyanappu30]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```

V1:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker tag quiz-app:v1 priyanappu30/quiz-app:v1
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker push priyanappu30/quiz-app:v1
The push refers to repository [docker.io/priyanappu30/quiz-app]
6e171a832c91: Layer already exists
461077a72fb7: Layer already exists
cda7f44f2bdd: Layer already exists
5005926c3003: Pushed
2e3f8db413f1: Pushed
37927ed901b1: Layer already exists
3697be50c98b: Layer already exists
43ce7e6786f1: Pushed
79b2f47ad444: Layer already exists
c6b30c3f1696: Layer already exists
3e6b9d1a9511: Layer already exists
e23f099911d6: Layer already exists
v1: digest: sha256:9e4c3eb0ccac110e307af86dff127b94e061b5a0e7228780e76c9e512f05bbe9 size: 856
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```

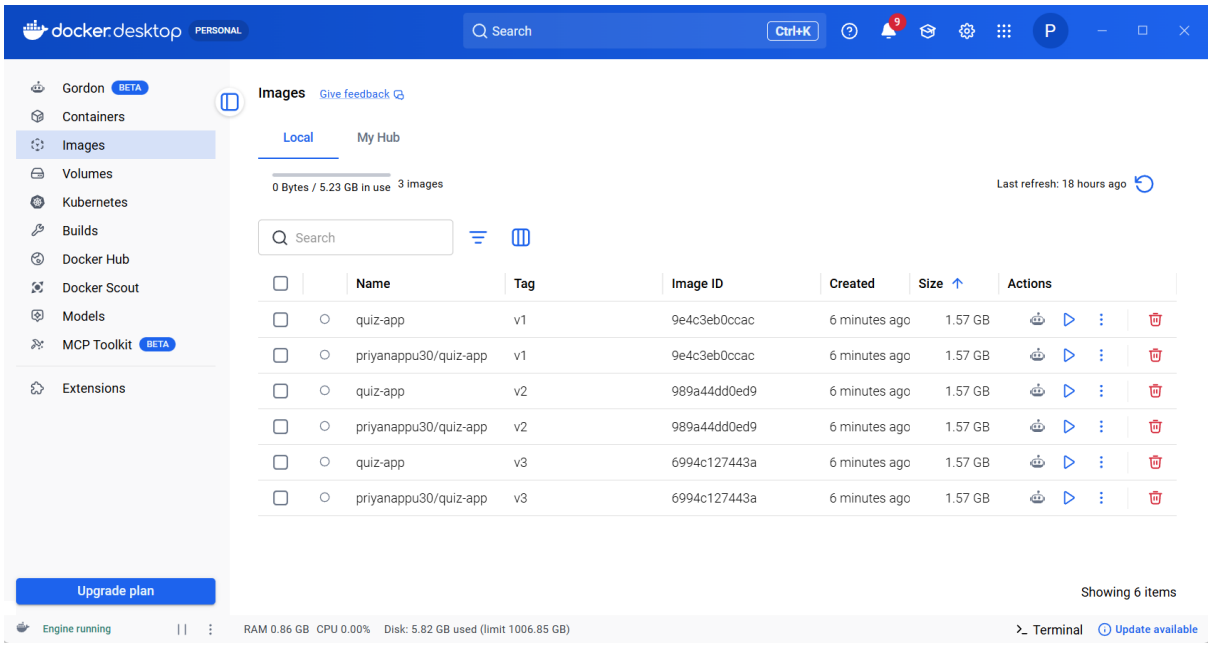
V2:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker tag quiz-app:v2 priyanappu30/quiz-app:v2
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker push priyanappu30/quiz-app:v2
The push refers to repository [docker.io/priyanappu30/quiz-app]
3e6b9d1a9511: Layer already exists
3697be50c98b: Layer already exists
461077a72fb7: Layer already exists
e23f099911d6: Layer already exists
37927ed901b1: Layer already exists
79b2f47ad444: Layer already exists
cda7f44f2bdd: Layer already exists
c6b30c3f1696: Layer already exists
43ce7e6786f1: Layer already exists
6e171a832c91: Layer already exists
5053453c20d5: Pushed
2e3f8db413f1: Layer already exists
v2: digest: sha256:989a44dd0ed9f2b5342fff1e02ea5121dca235c2e4f85a7ccf05f855c8992be6 size: 856
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```

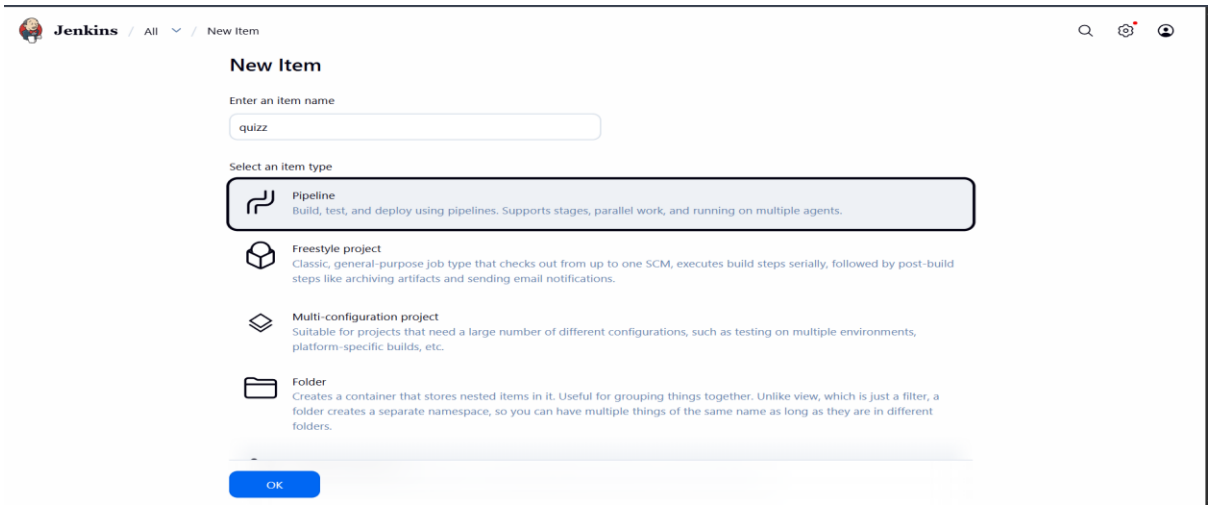
V3:

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker tag quiz-app:v3 priyanappu30/quiz-app:v3
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> docker push priyanappu30/quiz-app:v3
The push refers to repository [docker.io/priyanappu30/quiz-app]
79b2f47ad444: Layer already exists
37927ed901b1: Layer already exists
2e3f8db413f1: Layer already exists
cda7f44f2bdd: Layer already exists
3697be50c98b: Layer already exists
461077a72fb7: Layer already exists
6e171a832c91: Layer already exists
43ce7e6786f1: Layer already exists
2ef319bf7495: Pushed
3e6b9d1a9511: Layer already exists
e23f099911d6: Layer already exists
c6b30c3f1696: Layer already exists
v3: digest: sha256:6994c127443a1a63f2fdc70d19ea317ead5101b69c8ffa2aced74bcf507264e9 size: 856
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app>
```

Docker image :



STEP 8: Jenkins

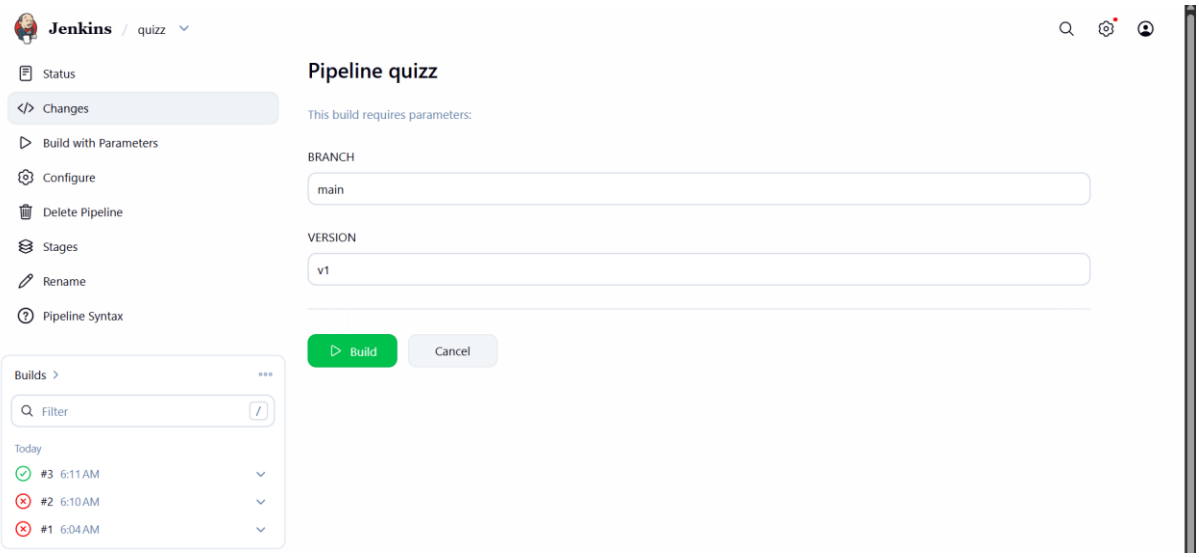


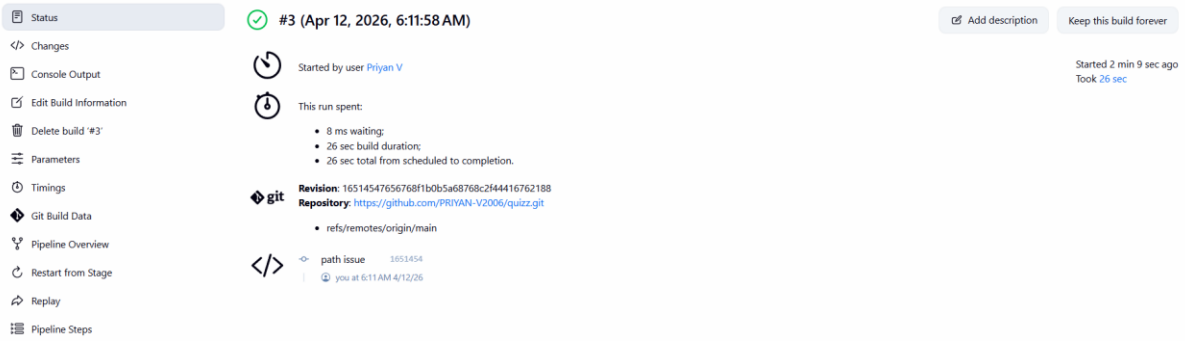
Pipeline Script

```
pipeline {
    agent any
```

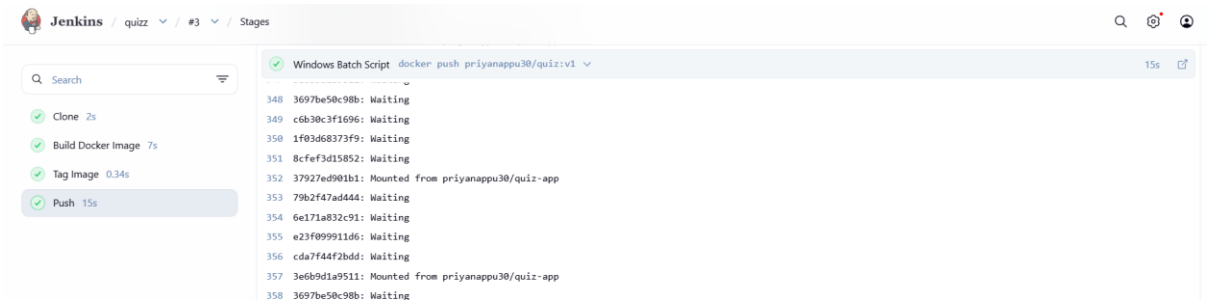
```
    parameters {
        string(name: 'BRANCH', defaultValue: 'main')
        string(name: 'VERSION', defaultValue: 'v1')
    }
    stages {
        stage('Clone') {
            steps {
                git branch: "${params.BRANCH}", url: '<your-github-repo>'
            }
        }
        stage('Build Docker Image') {
            steps {
                bat "docker build -t quiz:${params.VERSION} ."
            }
        }
        stage('Tag Image') {
            steps {
                bat "docker tag quiz:${params.VERSION}
yourdockerhub/quiz:${params.VERSION}"
            }
        }
        stage('Push') {
            steps {
                bat "docker push yourdockerhub/quiz:${params.VERSION}"
            }
        }
    }
}
```

V1:






```
8cfef3d15852: Pushed
70033b0a5b88: Pushed
1f03d68373f9: Pushed
v1: digest: sha256:0b0e57a2bf2d705cee2270ad9ffcffd0a9026b3ba05ec96535b38ae2e2f5bf55 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```



V2:

Jenkins / quizz / #7 / Parameters

Status
Changes
Console Output
Edit Build Information
Delete build '#7'
Parameters
Timings
Git Build Data
Pipeline Overview
Restart from Stage

Parameters

BRANCH

v2

VERSION

v2

Jenkins / quizz

Status
Changes
Build with Parameters
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax

Builds
Filter
Today
#4 6:16 AM
#3 6:11 AM

quizz

Add description

Permalinks

- Last build (#4), 50 sec ago
- Last stable build (#4), 50 sec ago
- Last successful build (#4), 50 sec ago
- Last failed build (#2), 7 min 34 sec ago
- Last unsuccessful build (#2), 7 min 34 sec ago
- Last completed build (#4), 50 sec ago

```
v2: digest: sha256:d0c387ae6bcd38187b754f9a671dc705c3a9de9a657061f2b9e4a399dbc9d5f8 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins / quizz / #4 / Stages

Search

Clone 2s
Build Docker Image 4s
Tag Image 0.35s
Push 10s

Windows Batch Script docker push priyanappu30/quiz:v2 10s

```
79b2f47ad444: Waiting
68b3baca5923: Waiting
e23f099911d6: Layer already exists
461077a72fb7: Waiting
8cfef3d15852: Waiting
1f03d68373f9: Waiting
c6b30c3f1696: Waiting
79b2f47ad444: Layer already exists
68b3baca5923: Waiting
461077a72fb7: Layer already exists
8cfef3d15852: Layer already exists
1f03d68373f9: Layer already exists
cda7f44f2bdd: Layer already exists
b862cd99a4ef: Waiting
b862cd99a4ef: Waiting
c6b30c3f1696: Layer already exists
68b3baca5923: Waiting
b862cd99a4ef: Waiting
68b3baca5923: Waiting
b862cd99a4ef: Waiting
68b3baca5923: Waiting
68b3baca5923: Pushed
b862cd99a4ef: Pushed
v2: digest: sha256:d0c387ae6bcd38187b754f9a671dc705c3a9de9a657061f2b9e4a399dbc9d5f8 size: 856
```

V3:

Jenkins / quizz / #8 / Parameters

Status
Changes
Console Output
Edit Build Information
Delete build '#8'
Parameters
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay

Parameters

BRANCH
v3

VERSION
v3

Jenkins / quizz / #8

Status
Changes
Console Output
Edit Build Information
Delete build '#8'
Parameters
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay

#8 (Apr 12, 2026, 6:26:31 AM)

Add description | Keep this build forever

Started by user Priyan V

Started 50 sec ago
Took 16 sec

This run spent:

- 10 ms waiting;
- 16 sec build duration;
- 16 sec total from scheduled to completion.

git
Revision: 3f741061ec6ec8fe8e48ee5f251e911b18e7c972
Repository: https://github.com/PRIYAN-V2006/quizz.git

- refs/remotes/origin/v3

Fix Dockerfile for v3 3f74186
you at 6:25 AM 4/12/26

```
v3: digest: sha256:2607078a0b33a9d31cf75be4ee5aaa26c5745807107cd0313db501a370ab98bb size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins / quizz / #8 / Stages

Search

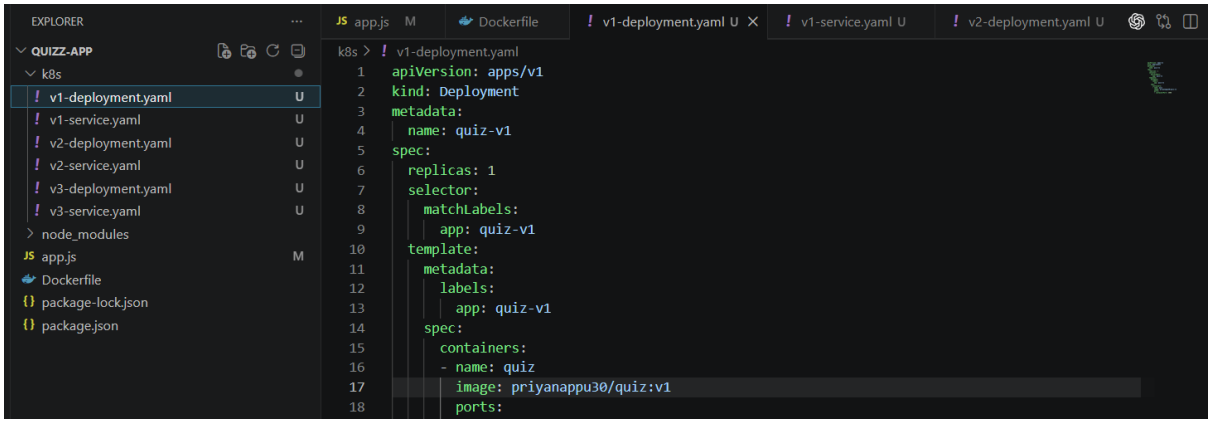
- Clone 2s
- Build Docker Image 2s
- Tag Image 0.37s
- Push 10s

Windows Batch Script `docker push priyanappu30/quiz:v3` 10s

```
291 6e171a832c91: Layer already exists
292 8cfef3d15852: Waiting
293 3697be50c98b: Waiting
294 3e6b9d1a9511: Waiting
295 8fb808c61622: Waiting
296 c0c4ad0d1c43: Waiting
297 461077a72fb7: Waiting
298 1f03d68373f9: Waiting
299 3e6b9d1a9511: Layer already exists
300 8fb808c61622: Waiting
301 c0c4ad0d1c43: Waiting
302 461077a72fb7: Layer already exists
303 1f03d68373f9: Layer already exists
304 79b2f47ad444: Layer already exists
305 8cfef3d15852: Layer already exists
306 3697be50c98b: Layer already exists
307 8fb808c61622: Waiting
308 c0c4ad0d1c43: Waiting
309 8fb808c61622: Waiting
310 c0c4ad0d1c43: Waiting
311 8fb808c61622: Waiting
312 c0c4ad0d1c43: Waiting
313 8fb808c61622: Waiting
314 c0c4ad0d1c43: Pushed
315 8fb808c61622: Pushed
316 v3: digest: sha256:2607078a0b33a9d31cf75be4ee5aaa26c5745807107cd0313db501a370ab98bb size: 856
```

STEP 9: KUBERNETES

Create ALL YAML FILES



Deploy everything

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> kubectl apply -f k8s/
service/quiz-v1-service created
service/quiz-v3-service created
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app> kubectl apply -f k8s/
service/quiz-v1-service unchanged
service/quiz-v2-service created
service/quiz-v3-service unchanged
```

```
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app\k8s> kubectl apply -f v1-deployment.yaml
>> kubectl apply -f v2-deployment.yaml
>> kubectl apply -f v3-deployment.yaml
deployment.apps/quiz-v1 created
error: no objects passed to apply
error: no objects passed to apply
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app\k8s> kubectl apply -f v1-deployment.yaml
>> kubectl apply -f v2-deployment.yaml
>> kubectl apply -f v3-deployment.yaml
deployment.apps/quiz-v1 unchanged
deployment.apps/quiz-v2 created
deployment.apps/quiz-v3 created
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app\k8s>
```

Verify

```
service/quiz-v3-service unchanged
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app\K8S> kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
quiz-v1-5b8b47b555-m9xr4    1/1     Running   0          80s
quiz-v2-599dcdb9c5-tcnc8    1/1     Running   0          80s
quiz-v3-59465c8cf-lh6sq     1/1     Running   0          80s
PS C:\Users\Priyan\OneDrive\Desktop\quizz-app\K8S>
```

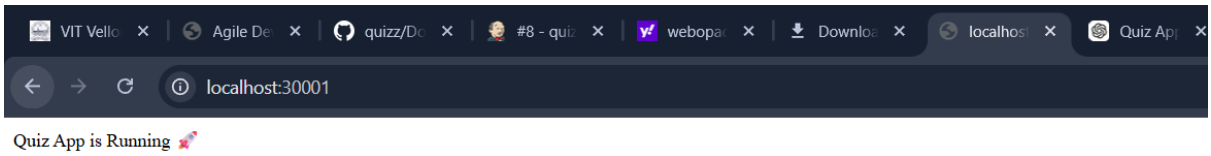
Kubernete Deployment in Local host

http://localhost:30001

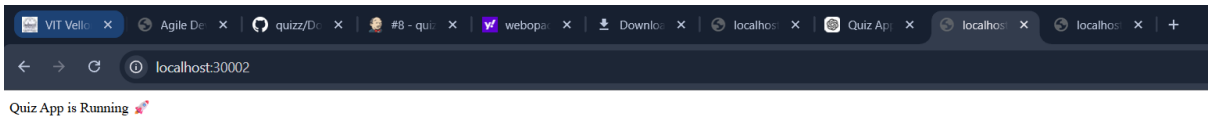
http://localhost:30002

http://localhost:30003

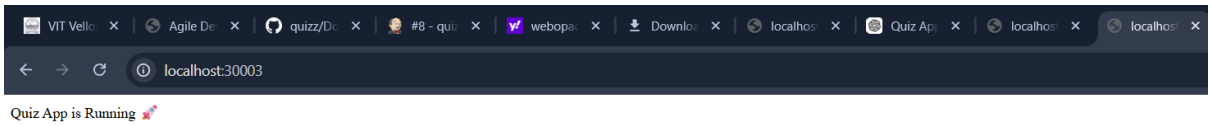
V1 :



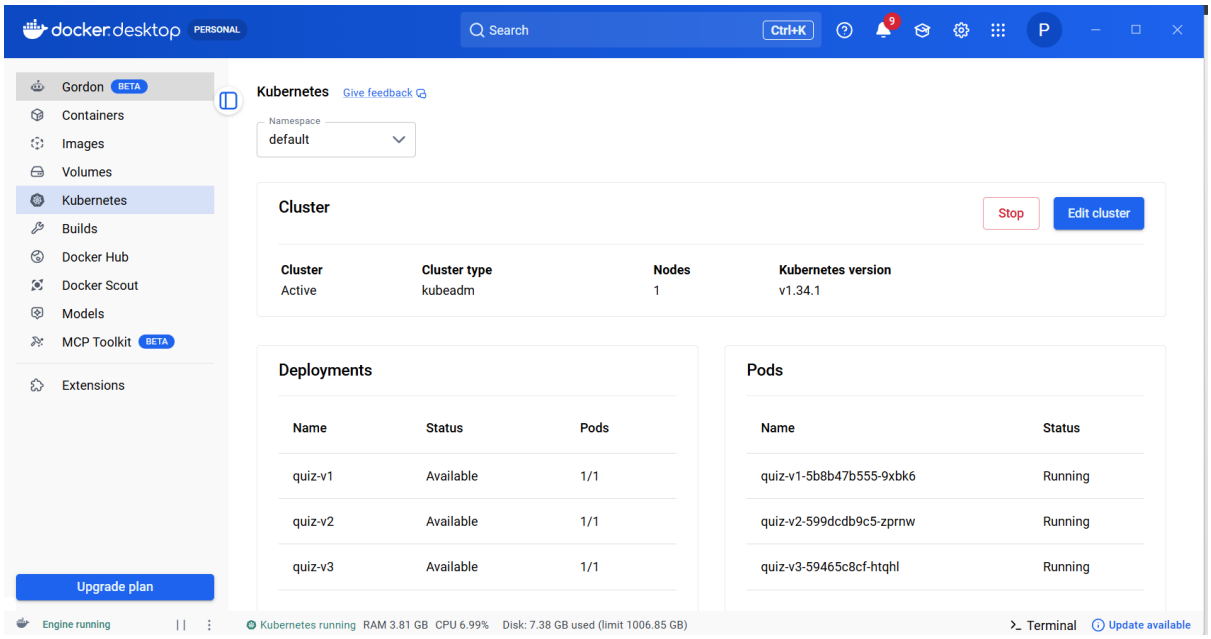
V2:



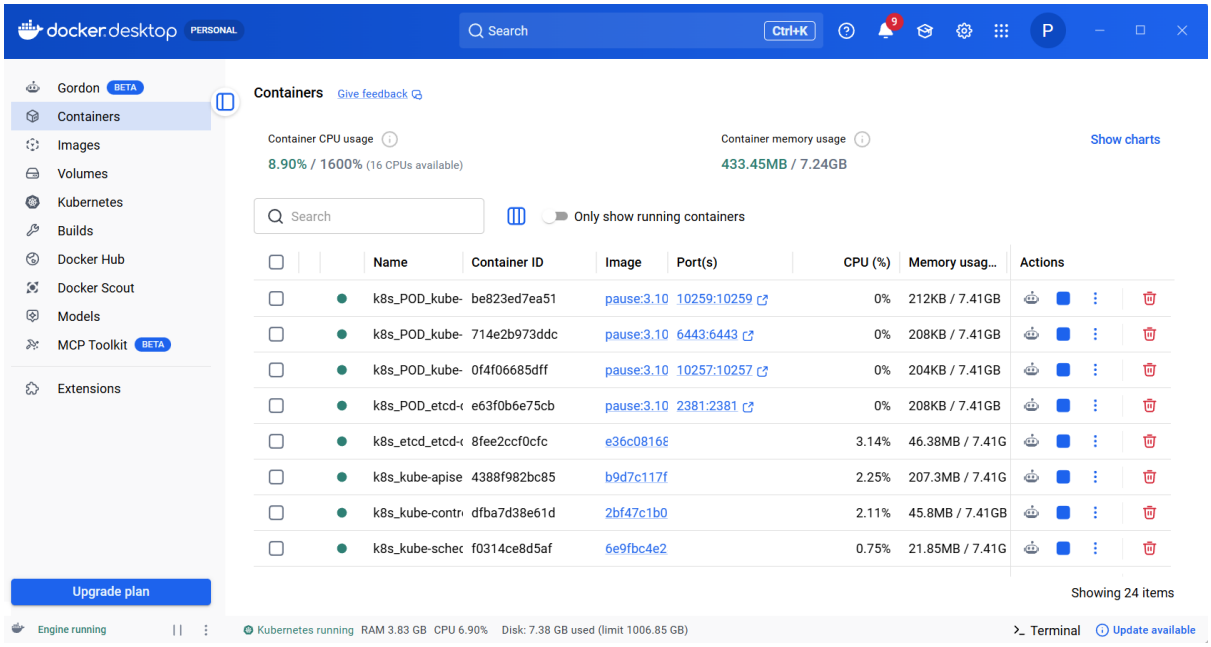
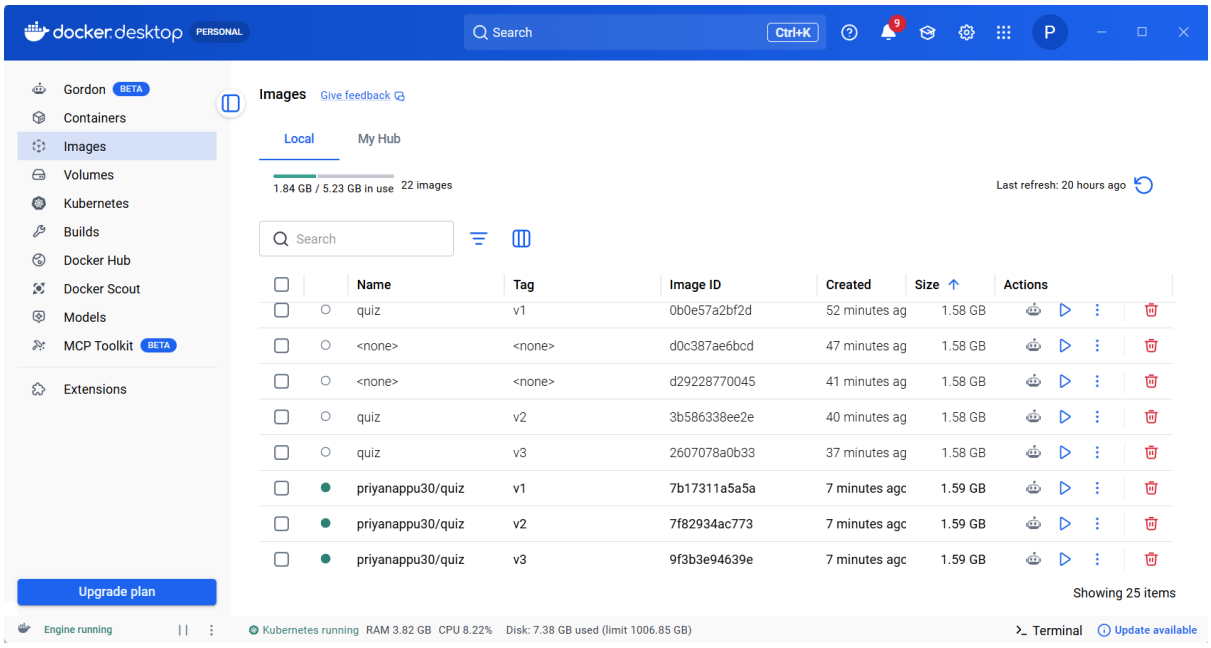
V3:



Docker Kubernetes



Docker container



Docker images